



Field	Details
Event	AtomQuest Hackathon 1.0
Conducted By	Atomberg Technologies
Problem	Goal Setting & Tracking Portal
Duration	48 Hours
Team Size	Solo
Approach	Vibe Coding (AI-assisted IDE)
App Name	GoalTrack
Primary Color	#FDB813 (Yellow) + #000000 (Black)

SECTION 1 — PROBLEM STATEMENT & UNDERSTANDING

1.1 The Core Problem

Companies today manage employee goals using spreadsheets, emails, and offline reviews. This creates:

- Managers cannot see team progress in real time
- Employees do not know how their work connects to company priorities
- HR has to manually piece together data during appraisals
- No accountability — things fall through the cracks

The Solution: Replace all of that with a clean, digital, role-based web portal covering the full goal lifecycle — creation, approval, quarterly tracking, and reporting.

1.2 The Three User Roles

Role	Who They Are	Core Responsibility
Employee	Any staff member	Creates goals, submits for approval, logs quarterly achievement
Manager (L1)	Direct line manager	Reviews and approves/edits/returns goals, reviews check-ins
Admin / HR	HR team or system admin	Controls windows, manages org hierarchy, exports reports

1.3 Key Concepts Explained

What is a Goal?

A goal is something an employee wants to achieve by end of year. It has a clear target, a timeframe, and a measurable metric.

Example: "Achieve Rs.40 Lakhs in regional sales revenue by 31st March 2026"

What is a KPI?

KPI (Key Performance Indicator) is the measuring stick for a goal. It answers: How do we know if you succeeded?

Employee Role	Example Goal	KPI Metric	UoM Type
Sales Rep	Achieve Rs.40L revenue	Revenue in Rupees	Numeric Min

HR Manager	Hire 20 new employees	Number of hires	Numeric Min
Customer Support	Reduce response time	Avg response in hours	Numeric Max
Safety Officer	Zero safety incidents	Number of incidents	Zero Based
Project Manager	Deliver project on time	Completion date	Timeline

1.4 Unit of Measurement (UoM) Types

UoM Type	When to Use	Score Formula	Example
Numeric (Min)	Higher actual = better	actual / target x 100	18L / 40L = 45%
Numeric (Max)	Lower actual = better	target / actual x 100 (capped 100%)	TAT: 5hrs target, 3hrs actual = 100%
Timeline	Complete by a date	On or before date = 100%, else 0%	Delivered by March 31 = 100%
Zero Based	Zero occurrences = success	actual = 0 = 100%, else 0%	0 safety incidents = 100%

1.5 What are Shared / Departmental Goals?

When a manager wants the whole team to share one goal, they "push" it to multiple employees at once.

- Goal title and target are FIXED — employees cannot change them
- Each employee can adjust only the WEIGHTAGE of the shared goal
- When the primary owner updates their achievement, it automatically SYNCs to all linked employees

Example: "Every sales rep must achieve Rs.40L revenue" — pushed to all 8 sales reps by the manager.

1.6 Quarterly Check-in Schedule

Period	Window Opens	Covers	What Happens
Goal Setting	May 1	Full year	Employees create & approve goals
Q1 Check-in	July 1	April – June	Employee logs actual vs planned
Q2 Check-in	October 1	July – September	Employee logs actual vs planned
Q3 Check-in	January 1	October – December	Employee logs actual vs planned
Q4 / Annual	March 1	Full year final	Final achievement capture

1.7 Evaluation Criteria

#	Criteria	What Judges Look For
1	Functionality	Does the full flow work without breaking?
2	BRD Adherence	Are ALL Phase 1 & Phase 2 requirements implemented correctly?
3	User Friendliness	Is the UI clean and intuitive for non-tech users?
4	Presence of Bugs	Edge cases handled? No broken flows?
5	Good-to-Have Features	Bonus features from Section 5 of BRD implemented?
6	Cost Optimisation	Infrastructure choices, API efficiency, caching, hosting cost

SECTION 2 — APP IDEA & HACKATHON STRATEGY

2.1 Our App Idea

A clean, modern, 3-role web portal for managing employee goals — from creation to approval to quarterly tracking — styled in black and yellow, with pre-loaded Atomberg-specific thrust areas and departments.

This replaces messy spreadsheets and email chains with a structured digital system that feels like it was built specifically for the judges company. That is the key differentiator.

2.2 Atomberg-Specific Customizations (Without Using Company Name/Logo)

Point	Customization	What We Do	Why It Works
2	Pre-loaded Thrust Areas	Pre-fill dropdown with real Atomberg department categories	Judges recognize their own departments instantly
3	Goal Suggestions	Suggest role-specific goals when thrust area is selected	Makes UX feel tailored and smart
5	Color Theme	Black (#000000) + Yellow (#FDB813) throughout	Judges see their own brand colors — instant familiarity

2.3 Pre-loaded Thrust Areas

#	Thrust Area	Relevant For	Example Goal
1	Sales & Revenue Growth	Sales reps, Area Managers	Achieve Rs.40L regional revenue
2	Product Innovation	R&D Engineers	Launch 3 new product features
3	Manufacturing Excellence	Plant / Production team	Zero defect rate below 0.5%
4	Marketing & Brand Building	Marketing team	Generate 500 qualified leads
5	Customer Delight	Support team	Achieve CSAT score of 90%
6	Supply Chain & Operations	Logistics, Procurement	Reduce delivery TAT to 2 days
7	People & Culture	HR team	Hire 20 engineers by Q2
8	Financial Discipline	Finance team	Reduce operating cost by 10%

2.4 Similar Apps in the Market

Tool	What It Does	Monthly Cost (1000 users)
Lattice	OKR tracking with quarterly cycles, 1:1 check-ins, reviews	Rs.6-9 Lakhs/month
Profit.co	Comprehensive OKR platform, planning + processes + people	Rs.6-9 Lakhs/month
15Five	Continuous performance management via weekly check-ins	Rs.5-8 Lakhs/month
Weekdone	Hierarchical OKRs with live dashboards and KPI monitoring	Rs.4-7 Lakhs/month
Betterworks	Goal setting, continuous feedback, team alignment	Rs.6-8 Lakhs/month
GoalTrack (Ours)	Purpose-built, in-house, tailored to this company	Rs.3,750/month (production)

Key insight: 56% of organizations still use spreadsheets. Your app is a purpose-built, zero-cost, in-house alternative that saves Rs.5-8 Lakhs every single month.

2.5 Hackathon Strategy — The Three Layer Approach

Layer	What	Priority
Layer 1	Core BRD — Phase 1 + Phase 2 completely and bug-free	NON-NEGOTIABLE
Layer 2	UI Polish — Black & yellow theme, loading states, empty states, responsive	NON-NEGOTIABLE
Layer 3	Analytics Module (bonus first), Escalation Module (bonus second)	If time permits

2.6 Bonus Feature Priority

Priority	Feature	Why This Choice
1st — Build this	Analytics Module	Visually impressive, charts visible instantly in demo, no external dependencies, uses only Recharts
2nd — If time permits	Escalation Module	Practical business value, shows real thinking, harder to demo live but explainable
Skip	Azure AD SSO	Complex setup, can break during demo, eats 3-4 hours of build time
Skip	Teams Integration	Requires Microsoft tenant, too risky for solo 48hr build

SECTION 3 — COMPLETE APP WORKFLOW

3.1 Employee Journey — Step by Step

Step 1: Goal Creation Inputs (6 Fields)

#	Field	Input Type	Rules / Notes
1	Thrust Area	Dropdown	Required — pre-loaded with 8 Atomberg categories
2	Goal Title	Text (max 200 chars)	Required — short descriptive title
3	Description	Textarea	Required — explains the goal in detail
4	Unit of Measure	Dropdown	Required — numeric_min, numeric_max, timeline, zero_based
5	Target	Number or Date picker	Required — switches dynamically based on UoM selected
6	Weightage (%)	Number input	Required — min 10%, all goals must total exactly 100%

Step 2: Goal Validation Rules (All enforced in real time)

- Total weightage of all goals MUST equal exactly 100% before submission
- Each individual goal MUST have minimum 10% weightage
- Maximum 8 goals per employee per cycle
- Cannot submit for approval if total weightage is not 100%
- WeightageBar updates live as employee types the weightage value

Step 3: Quarterly Check-in Inputs

#	Field	Input Type	Required?
1	Actual Achievement	Number or Date (matches UoM)	Yes — cannot submit without this
2	Status	Dropdown: Not Started / On Track / Completed	Yes
3	Notes	Textarea	No — optional context for manager (our addition)

3.2 Manager Journey — Step by Step

Goal Review Actions per Goal

- Approve as-is — goal status immediately becomes locked
- Edit inline — change target or weightage, then approve (triggers audit log automatically)
- Return for rework — with mandatory comment explaining why; employee can then edit and resubmit
- Approve All — approves and locks all goals for that employee in one click

Quarterly Check-in Review

- View Planned vs Actual side by side for every goal in a table
- See auto-computed progress score per goal (calculated by DB trigger)
- Add structured check-in comment to document the 1:1 discussion
- Mark entire check-in as complete — tracked for completion dashboard

3.3 Admin Journey — Step by Step

Cycle & Window Management

- Create a new yearly cycle with all window dates
- Open or close any window manually at any time
- Only ONE cycle can be active at a time

User Management

- Create new users and assign role (employee / manager / admin)
- Assign manager_id to link employees to their manager
- Edit department and reporting structure

Shared Goals

- Create a departmental KPI with fixed title and target
- Select multiple employees to push it to
- Employees receive the goal with locked title/target — can only change weightage
- Primary owner achievement update syncs to all linked employees automatically

Exception Handling

- Unlock a goal after it is locked (creates audit log entry automatically)
- Reassign goals if someone changes teams

Reports & Oversight

- Export Achievement Report as CSV or Excel
- View real-time Completion Dashboard — who has/has not done check-ins
- View full Audit Trail — every post-lock change with who/what/when
- Configure Escalation Rules — N days threshold per rule type

3.4 Complete Bird's Eye Flow

```
MAY 1 - Goal Setting Window Opens
|
v
Employee creates goals (max 8, min 10% each, total = 100%)
|
v
Employee submits for Manager Approval
|
v
Manager reviews -> Approves / Edits / Returns
| (if returned, employee edits and resubmits)
v
Goals LOCKED after full approval
|
v
JULY - Q1 Check-in Window Opens
|
v
Employee logs Actual Achievement + Status + Notes
|
v
DB Trigger auto-calculates Progress Score
|
v
Manager reviews Planned vs Actual + adds comment
|
v
Repeat each quarter: October (Q2), January (Q3)
|
v
MARCH - Q4 / Annual Final Check-in
|
v
Admin exports Achievement Report -> Performance Review Ready
```

SECTION 4 — DATABASE DESIGN (4-STEP PROCESS)

4.1 Step 1 — Requirement Analysis

Entities Identified

Entity	Source in BRD	Description
USER	Section 3 — User Roles	All 3 roles in one table, self-referencing for org hierarchy
GOAL	Section 2.1 — Goal Creation	Core entity — one row per goal per employee per cycle
THRUST AREA	Section 2.1 — Thrust Area field	Pre-loaded category lookup table
CYCLE	Section 2.3 — Check-in Schedule	Controls all window open/close dates
CHECK-IN	Section 2.2 — Achievement Tracking	One row per goal per quarter — stores actual achievement
SHARED GOAL	Section 2.1 — Departmental KPI	Junction table for M:N relationship between goals and users
AUDIT LOG	Section 4 — Reporting	Every post-lock change automatically logged
ESCALATION	Section 5.3 — Escalation Module	Triggered instances of escalation rules
ESCALATION RULE	Section 5.3 — Escalation Module	Admin-configured N-day thresholds per rule type

Relationships Identified (11 total)

Relationship	Entity 1	Cardinality	Entity 2
Creates	User (employee)	1 to Many	Goal
Approves	User (manager)	1 to Many	Goal
Belongs to	Goal	Many to 1	Thrust Area
Belongs to	Goal	Many to 1	Cycle
Has	Goal	1 to Many	Check-in
Reviews	User (manager)	1 to Many	Check-in
Shared to	Goal	Many to Many	User (junction table needed)
Has	Goal	1 to Many	Audit Log
Makes	User	1 to Many	Audit Log

Has	User	1 to Many	Escalation
Triggers	Escalation Rule	1 to Many	Escalation

Business Constraints (13 Rules Extracted from BRD)

1. Total weightage across all goals = exactly 100%
2. Minimum weightage per goal = 10%
3. Maximum goals per employee = 8
4. Goals locked after manager approval — no edits without Admin
5. Shared goals: employee can only change weightage
6. Shared goals: title and target are read-only
7. Achievement by primary owner syncs to all linked employees
8. Employee can only update during open window
9. One check-in per goal per quarter
10. Progress score auto-calculated based on UoM type
11. Only one active cycle at a time
12. Admin controls all window open/close dates
13. Every post-lock change must log who/what/when

4.2 Step 2 — Logical Design (ER Diagram)

ER Mapping Rules Applied:

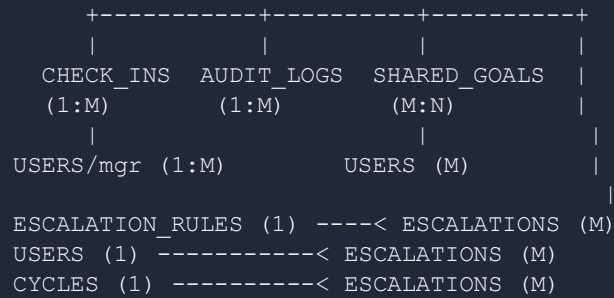
Rule	Applied Where
Rule 1 — Entity = Table	All 9 entities become 9 tables
Rule 2 — Attribute = Column	All attributes become typed columns
Rule 3 — PK stays PK	All UUID primary keys preserved
Rule 4 — 1:M = FK on Many side	employee_id in goals, goal_id in check_ins, etc.
Rule 5 — M:N = Junction table	SHARED_GOALS table created for Goal <-> User M:N
Rule 6 — Self-reference	manager_id in USERS references users.id

Entity-Relationship Summary Diagram

```

THRUST_AREAS (1) ----< GOALS (M)
                        |
CYCLES (1) -----< GOALS (M)
                        |
USERS/employee (1) ---< GOALS (M)
                        |
USERS/manager (1) ----< GOALS (M)
                        |

```



4.3 Step 3 — Physical Design (Supabase Optimization)

Complete Database Schema — All 9 Tables

Table 1: users

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	Linked to Supabase auth.users
name	VARCHAR(100)	NOT NULL	
email	VARCHAR(255)	NOT NULL UNIQUE	
role	ENUM('employee','manager','admin')	NOT NULL	Controls access everywhere
department	VARCHAR(100)		Atomberg department name
manager_id	UUID	FK -> users.id	Self-referencing for org hierarchy
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Table 2: thrust areas

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
name	VARCHAR(100)	NOT NULL UNIQUE	Pre-seeded with 8 Atomberg areas
description	TEXT		Optional longer description
is_active	BOOLEAN	DEFAULT true	Admin can deactivate

created_at	TIMESTAMPTZ	DEFAULT NOW()	
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Table 3: cycles

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
name	VARCHAR(100)	NOT NULL	e.g. 'FY 2025-26'
goal_setting_start/end	DATE	NOT NULL	Goal setting window
q1_start / q1_end	DATE	NOT NULL	Q1 window
q2_start / q2_end	DATE	NOT NULL	Q2 window
q3_start / q3_end	DATE	NOT NULL	Q3 window
q4_start / q4_end	DATE	NOT NULL	Q4 window
is_active	BOOLEAN	DEFAULT false	Partial unique index — only one active
created_by	UUID	FK -> users.id	Admin who created this cycle

Table 4: goals

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
employee_id	UUID	FK -> users.id NOT NULL	Goal owner
manager_id	UUID	FK -> users.id	Approving manager
thrust_area_id	UUID	FK -> thrust_areas.id	
cycle_id	UUID	FK -> cycles.id	
title	VARCHAR(200)	NOT NULL	
description	TEXT		
uom_type	ENUM(4 types)	NOT NULL	numeric_min/max/timeline/zero_based
target_value	DECIMAL(15,2)		For numeric and zero_based
target_date	DATE		For timeline goals only
weightage	DECIMAL(5,2)	NOT NULL CHECK >= 10	Min 10% per goal
status	ENUM(5 states)	DEFAULT 'draft'	draft/submitted/approved/returned/locked

is_shared_goal	BOOLEAN	DEFAULT false	
manager_comment	TEXT		Return reason or edit note
created_at / updated_at	TIMESTAMPTZ	DEFAULT NOW()	updated_at via trigger

Table 5: check_ins

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
goal_id	UUID	FK -> goals.id NOT NULL	
employee_id	UUID	FK -> users.id NOT NULL	
manager_id	UUID	FK -> users.id	
cycle_id	UUID	FK -> cycles.id	
quarter	ENUM(Q1/Q2/Q3/Q4)	NOT NULL	
actual_value	DECIMAL(15,2)		For numeric and zero_based
actual_date	DATE		For timeline goals
status	ENUM(3 states)	NOT NULL	not_started/on_track/completed
notes	TEXT		Optional employee note — our addition
progress_score	DECIMAL(5,2)	CHECK 0-100	Auto by DB trigger — never set manually
manager_comment	TEXT		Manager check-in comment
manager_checked_in	BOOLEAN	DEFAULT false	Has manager reviewed?
created_at / updated_at	TIMESTAMPTZ	DEFAULT NOW()	updated_at via trigger

UNIQUE CONSTRAINT: (goal_id, quarter, cycle_id) — enforces one check-in per goal per quarter.

Table 6: shared_goals (Junction Table — M:N)

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
original_goal_id	UUID	FK -> goals.id	The master/source goal
pushed_to_user_id	UUID	FK -> users.id	Recipient employee

pushed_by_user_id	UUID	FK -> users.id	Manager/admin who pushed
weightage	DECIMAL(5,2)	NOT NULL CHECK >= 10	Only field recipient can change
created_at	TIMESTAMPTZ	DEFAULT NOW()	

UNIQUE CONSTRAINT: (original_goal_id, pushed_to_user_id) — same goal cannot be pushed to same employee twice.

Table 7: audit_logs

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
goal_id	UUID	FK -> goals.id	
changed_by	UUID	FK -> users.id	Who made the change
change_type	VARCHAR(100)	NOT NULL	e.g. 'target_changed', 'goal_unlocked'
old_value	JSONB		Flexible — stores any changed field as JSON
new_value	JSONB		Flexible — stores new value as JSON
reason	TEXT		Why was the change made
created_at	TIMESTAMPTZ	DEFAULT NOW()	Auto-created by DB trigger

Table 8: escalation_rules

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
type	ENUM(3 types)	NOT NULL UNIQUE	goal_not_submitted/goal_not_approved/checkin_not_completed
days_threshold	SMALLINT	NOT NULL	N days before escalation triggers
is_active	BOOLEAN	DEFAULT true	Admin can toggle on/off
created_by	UUID	FK -> users.id	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last time rule was changed

Table 9: escalations

Column	Data Type	Constraint	Notes
id	UUID	PRIMARY KEY	
rule_id	UUID	FK -> escalation_rules.id	
triggered_for	UUID	FK -> users.id	Who needs to take action
triggered_against	UUID	FK -> users.id	Who is being escalated about
cycle_id	UUID	FK -> cycles.id	
type	ENUM(3 types)	NOT NULL	Same as rule types
quarter	ENUM(Q1-Q4)	NULLABLE	Null for goal setting escalations
status	ENUM(3 states)	DEFAULT 'pending'	pending/notified/resolved
escalation_level	SMALLINT	DEFAULT 1 CHECK 1-3	1=employee, 2=manager, 3=HR
created_at	TIMESTAMPTZ	DEFAULT NOW()	When triggered
resolved_at	TIMESTAMPTZ	NULLABLE	When resolved

4.4 Progress Score Formulas (DB Trigger + Live Preview)

```
-- DB Trigger fires on check_in INSERT or UPDATE

IF uom_type = numeric_min THEN
    score = LEAST((actual_value / target_value) * 100, 100)

ELSIF uom_type = numeric_max THEN
    score = LEAST((target_value / actual_value) * 100, 100)

ELSIF uom_type = timeline THEN
    IF actual_date <= target_date THEN score = 100
    ELSE score = 0

ELSIF uom_type = zero_based THEN
    IF actual_value = 0 THEN score = 100
    ELSE score = 0

-- Always round to 2 decimal places
-- Always handle division by zero gracefully
```

4.5 Three Database Triggers

Trigger	Fires When	Does What
trigger_progress_score	check_ins INSERT or UPDATE	Auto-calculates and stores progress_score based on UoM formula
trigger_audit_log	goals UPDATE (when status = locked)	Auto-creates audit_log entry with old_value, new_value, changed_by
trigger_updated_at	goals or check_ins UPDATE	Auto-updates the updated_at timestamp column

4.6 Indexes for Performance

Index Name	Table.Column	Speeds Up
idx_goals_employee_id	goals.employee_id	Employee fetching their own goals
idx_goals_cycle_id	goals.cycle_id	Fetching all goals in a cycle
idx_goals_manager_id	goals.manager_id	Manager fetching team goals
idx_checkins_goal_id	check_ins.goal_id	Fetching check-ins for a goal
idx_checkins_quarter	check_ins.quarter	Filtering by quarter
idx_goals_status	goals.status	Admin filtering by status
idx_audit_goal_id	audit_logs.goal_id	Audit log lookup per goal
idx_escalations_user	escalations.triggered_for	Escalation lookup per user
idx_thrust_areas_active	thrust_areas.is_active	Active thrust areas dropdown

4.7 Row Level Security (RLS) Policies

Table	Employee Policy	Manager Policy	Admin Policy
users	See own profile only	See own profile + direct reports	See all users
goals	See/edit own goals only	See team goals (read)	Full access
check_ins	See/edit own check-ins	See team check-ins (read + comment)	Full access
audit_logs	No access	No access	Full read access
escalations	No access	No access	Full access
cycles	Read active cycle	Read active cycle	Full CRUD
thrust_areas	Read active only	Read active only	Full CRUD

4.8 Supabase Realtime Configuration

Table	Realtime?	Why
goals	YES	Manager approves => employee sees status change instantly
check_ins	YES	Manager adds comment => employee sees it live
audit_logs	YES	Admin sees all post-lock changes in real time
escalations	YES	Admin sees new escalations without refresh
users	NO	Changes rarely — no realtime needed
cycles	NO	Admin changes occasionally — no realtime needed
thrust_areas	NO	Static data — cached forever

SECTION 5 — TECH STACK & ARCHITECTURE

5.1 Final Tech Stack

Strategy: Build fast with Supabase + Vercel for demo. Present Azure as production path in mandatory architecture diagram.

Layer	Technology	Why This Choice
Framework	Next.js 14 (App Router)	AI tools generate it cleanly, built-in routing, server + client components
Styling	Tailwind CSS	Black & yellow theme in minutes, utility-first, no raw CSS
Components	shadcn/ui	Beautiful pre-built components (Button, Card, Table, Dialog, Badge)
Database	Supabase (PostgreSQL)	Free tier generous, built-in auth, realtime, visual dashboard
Authentication	Supabase Auth + RLS	3-role auth + row-level security in one tool
Real-time	Supabase Realtime	Completion dashboard live updates, goal approval notifications
Audit Trail	PostgreSQL trigger + table	Automatic, reliable, cannot be bypassed by frontend
Excel Export	SheetJS (xlsx)	Mandatory BRD requirement, well-supported, AI knows it well
Charts	Recharts	Analytics bonus feature, simple API, React-native
Email	Resend	Free tier 100/day, simple API, AI generates it easily
Deployment	Vercel	One-click Next.js deploy, free tier, instant live URL in 2 minutes
Version Control	GitHub	Mandatory submission requirement

5.2 Hackathon vs Production Architecture

Layer	Hackathon Stack (What We Build)	Production Stack (Architecture Diagram)
Framework	Next.js 14	Next.js 14 (same)
Database	Supabase PostgreSQL	Azure SQL Database (Indian region)
Authentication	Supabase Auth	Microsoft Entra ID (Azure AD SSO)

Hosting	Vercel	Azure App Service (Indian region)
Caching	Next.js built-in cache	Azure Cache for Redis
Email/Notifications	Resend	Microsoft Exchange + Teams Bot
File Storage	Supabase	Azure Blob Storage
Monthly Cost	Rs.0 (free tiers)	~Rs.3,750/month (~\$45/month)
vs Lattice/Profit.co	-	Rs.6-9 Lakhs/month saved

5.3 Cost Optimization — Full Analysis

Criteria	Our Approach	Evidence
Infrastructure Choices	Serverless on Vercel — pay per use	Free tier covers demo; scales on demand
API Call Efficiency	Joined queries, no N+1, select only needed columns	e.g. one query fetches employee + goals + thrust area name
Caching — Static	next: { revalidate: false } for thrust areas	Fetches once, cached permanently until admin changes
Caching — Semi-static	next: { revalidate: 3600 } for cycle dates	Cycle dates change rarely — 1hr cache is perfect
Caching — Dynamic	next: { revalidate: 300 } for goals/check-ins	5min cache, fresh enough for most users
Caching — Real-time	cache: 'no-store' for completion dashboard	Realtime data must never be stale
Hosting Cost Awareness	\$0 demo, \$45/month production, \$8000+/month competitor	Clear ROI story for judges

SECTION 6 — UI/UX DESIGN PLAN & SCREEN LAYOUTS

6.1 Design System

Element	Color	Hex	Used For
Primary	Yellow	#FDB813	All primary buttons, progress bars, active sidebar links, accents
Primary Hover	Dark Yellow	#E5A510	Button hover states
Sidebar/Navbar	Black	#000000	All sidebars and top navigation bars
Page Background	Light Grey	#F5F5F5	All page backgrounds
Cards	White	#FFFFFF	All content cards and panels
Success	Green	#22C55E	Approved status badges, 100% weightage indicator
Warning	Amber	#F59E0B	Submitted/pending badges, partial progress
Danger	Red	#EF4444	Returned goals, error states, over-100% weightage
Muted	Dark Grey	#6B7280	Locked goals, disabled states, secondary text
Text Primary	Near Black	#111827	All body text
Border	Light Grey	#E5E7EB	All card and table borders

6.2 Status Badge Color Reference

Status	Color	Hex	When Used
Draft	Grey	#6B7280	Goal created but not submitted
Submitted	Yellow	#FDB813	Submitted to manager, awaiting review
Approved	Green	#22C55E	Manager approved, goal is now locked
Returned	Red	#EF4444	Manager returned — employee must rework
Locked	Dark Grey	#374151	Approved and locked — no edits allowed
Not Started	Grey	#6B7280	Check-in status — not yet begun
On Track	Yellow	#FDB813	Check-in status — in progress
Completed	Green	#22C55E	Check-in status — fully done

6.3 All 14 Screens — Detailed Layouts

Screen 1 — Login Page (All Roles)

```

+-----+
|           [ GoalTrack ] (yellow text)           |
|-----+
| | Email Address | | |
| | [ _____ ] | | |
| | Password | | |
| | [ _____ ] | | |
| | [##### Sign In #####] (yellow btn) | | |
| | Error: Invalid credentials (red, if any) | | |
| |-----+
| | Note: No registration - Admin creates accounts |
|-----+
Background: #F5F5F5 | Card: White | Logo: Black bar

```

Flow after login: Employee => /employee/dashboard | Manager => /manager/dashboard | Admin => /admin/dashboard

Screen 2 — Employee Dashboard

```

+-----+
| [GoalTrack] | FY 2025-26 | Q1 Check-in: OPEN | Logout |
| (black bar) +-----+
| | Welcome back, Rahul S! |
| SIDEBAR |
| (black bg) | +-----+ +-----+ +-----+ +-----+
| > Dashboard | | Total | |Approved| |Pending | |Weight||
| My Goals | | Goals | | Goals | |Approval| | 100% ||
| Check-ins | | 5 | | 3 | | 2 | | OK ||
| | +-----+ +-----+ +-----+ +-----+
| |
| | Goals Summary | | | | |
| | +---Title-----+---Weight+---Status---+ |
| | | Revenue Rs40L | 30% | Approved | |
| | | New Partners | 20% | Pending | |
| | | Zero Incidents| 10% | Returned | |
| | +-----+ +-----+ +-----+ +-----+
| |
| | [### + Add New Goal ###] (yellow btn) |
|-----+

```

Screen 3 — My Goals Page (Employee)

```

+-----+-----+
| SIDEBAR | My Goals — FY 2025-26 | | | |
|         | [### + Add Goal ###] [Submit All for Appr] |
|         | |
|         | Weightage: [#####] 85/100% |
|         | Warning: Need 15% more to reach 100% |
|         | |
|         | +-----+ |
|         | | Achieve Rs.40L Regional Revenue | |
|         | | Thrust: Sales & Revenue Growth | |
|         | | UoM: Numeric (Min) | Target: Rs.40,00,000 | |
|         | | Weightage: 30% | Status: APPROVED (G) | |
|         | | [View] | |
|         | +-----+ |
|         | +-----+ |
|         | | Onboard 10 New Channel Partners | |
|         | | Thrust: Sales & Revenue Growth | |
|         | | UoM: Numeric (Min) | Target: 10 | |
|         | | Weightage: 20% | Status: RETURNED (R) | |
|         | | Manager Note: "Target too low, revise" | |
|         | | [Edit & Resubmit] | |
|         | +-----+ |
+-----+-----+
G=Green badge R=Red badge Y=Yellow badge

```

Screen 4 — Create / Edit Goal Form (Employee)

```

+-----+-----+
| Create New Goal | [X] Close |
+-----+-----+
| Thrust Area * |
| [Sales & Revenue Growth v] |
| |
| Goal Title * |
| [Achieve Rs.40L regional revenue ] |
| |
| Description |
| [Drive sales across South Zone by... ] |
| [onboarding new distributors... ] |
| |
| Unit of Measure * |
| [Numeric (Min) — Higher is better v] |
| |
| Target Value * | (shows date picker |
| [40,00,000 ] | if UoM = timeline) |
| |
| Weightage (%) * |
| [30 ] Remaining: 45% | Min: 10% |
| ! Error if < 10% or > remaining available |
| |
| [Cancel] | [### Save Goal ###] (yellow) |
+-----+-----+

```

Screen 5 — Quarterly Check-in Page (Employee)

```

+-----+-----+
| SIDEBAR | Q1 Check-in — April to June 2025 | | |
|         | Window: Open until 31st July 2025 |
|         |                                     |
|         | Overall Q1 Score: 62%             |
|         | [#####-----] 62%              |
|         |                                     |
|         | +-----+                         |
|         | | Achieve Rs.40L Regional Revenue | |
|         | | Target: Rs.40,00,000            | |
|         | | -----                        | |
|         | | Actual Achievement: [Rs. 18,00,000 ] | |
|         | | Status: [On Track          v]    | |
|         | | Notes: [Revenue improving in Q1... ] | |
|         | |           (Optional — context for manager) | |
|         | |                                     | |
|         | | Projected Score: 45%   (updates live) | |
|         | +-----+                         |
|         |                                     |
|         | [Save Progress]  [## Submit Check-in ##] |
+-----+-----+
Note: If window closed => show "Next window opens Oct 1"

```

Screen 6 — Manager Dashboard

```

+-----+-----+
| [GoalTrack] | Manager Dashboard — FY 2025-26 | | | | | | | | |
| SIDEBAR    | +-----+                         |
| > Dashboard | +-----+ +-----+ +-----+ +-----+ |
| Team Goals | | Team   | |Pending | |Check-in| |Check-in| |
| Checkin    | |Members | |Approval| |Complete| |Pending | |
| Shared     | | 8      | | 3      | | 5      | | 3      | |
| Goals      | +-----+ +-----+ +-----+ +-----+ |
|             |                                     |
|             | Team Overview                       |
|             | +---Name-----+---Goals---+---Status---+---Q1---+ |
|             | | Rahul S      | 5/5      | Approved  | Done   | |
|             | | Priya M      | 3/5      | Pending   | Pending| |
|             | | Arun K       | 5/5      | Returned  | Pending| |
|             | | Sneha P      | 0/0      | Not Started| -      | |
|             | +-----+ +-----+ +-----+ +-----+ |
+-----+-----+

```

Screen 7 — Goal Review Page (Manager — per employee)

```

+-----+-----+
| SIDEBAR    | Reviewing Goals — Rahul Sharma | |
|             | Department: Sales | Submitted: 12 May 2025 |
|             | Total Weightage: 100% OK       |
+-----+-----+

```



```

| | +-----+ | | |
| | Achieve Rs.40L Regional Revenue | |
| | UoM: Numeric (Min) | Target: Rs.40,00,000 | |
| | Weightage: 30% | |
| | ----- | |
| | Edit Target: [Rs. 40,00,000 ] | |
| | Edit Weightage: [30 ]% | |
| | Comment: [ ] | |
| | | | |
| | [Return] [Edit & Approve] [### Approve ###] | |
| | +-----+ |
| | | |
| | [Return All] [### Approve All Goals ###] |
| | +-----+ |

```

Screen 8 — Check-in Review Page (Manager — per employee)

SIDEBAR				
Q1 Check-in Review - Rahul Sharma				
Planned vs Actual				
Goal	Planned	Actual	Score	
Revenue	Rs.40L	Rs.18L	45%	
Partners	10	4	40%	
Incidents	0	0	100%	
Overall Q1 Score: 61.7%				
[#####]				
Manager Check-in Comment *				
[Discussed pipeline. Revenue on track.]				
[Q2 acceleration plan agreed.]				
[### Submit Check-in Review ###] (yellow)				

Screen 9 — Push Shared Goal Page (Manager/Admin)

SIDEBAR	Push Departmental KPI to Team	
	Goal Details (recipients cannot edit these)	
	+-----+	
	Thrust Area:	Sales & Revenue Growth
	Goal Title:	Achieve Rs.40L Revenue
	Description:	[]
	UoM:	Numeric (Min)
	Target:	Rs.40,00,000 (LOCKED)
	+-----+	

```
| | Select Recipients | |
```

[x] Rahul Sharma	[x] Priya Menon
[x] Arun Kumar	[] Sneha Patel
[Select All]	[Deselect All]

```
| | [### Push Goal to Selected Employees ###] | |
```

Screen 10 — Admin Dashboard

```

+-----+
| [GoalTrack] | Admin Dashboard - FY 2025-26 |
| SIDEBAR    | +-----+
| > Dashboard | +-----+ +-----+ +-----+ +-----+
|   Cycles   | | Total   | | Goals   | | Check-in | | Escalat. |
|   Users    | | Employs | | Submit  | | Complet. | | Trigger  |
|   Reports  | | 124     | | 89%     | | 72%     | | 5        |
|   Audit    | +-----+ +-----+ +-----+ +-----+
|   Escalate |
|             | Department Completion (this quarter)
|             | Sales      [#####-----] 80%
|             | R&D        [#####-----] 50%
|             | Manufacturing[#####] 100%
|             | Marketing  [#####-----] 70%
|             |
|             | Recent Activity
|             | - Rahul S submitted goals (2 mins ago)
|             | - Priya M check-in done (1 hour ago)
|             | - Escalation triggered: Arun K (3 hrs ago)
+-----+

```

Screen 11 — Cycle Management Page (Admin)

Cycle Management				
Current: FY 2025-26 [GREEN: Active]				
Window	Opens	Closes	Status	Action
Goal Setting	May 1	May 31	Closed	Edit
Q1 Check-in	Jul 1	Jul 31	OPEN(G)	Close
Q2 Check-in	Oct 1	Oct 31	Upcoming	Edit
Q3 Check-in	Jan 1	Jan 31	Upcoming	Edit
Q4 / Annual	Mar 1	Apr 30	Upcoming	Edit
[### + Create New Cycle ###] (yellow)				

G=Green badge. Status: Upcoming=grey, Open=green, Closed=red

Screen 12 — Reports Page (Admin — 3 Tabs)

```

+-----+-----+
| SIDEBAR | Reports & Analytics |
|         | [Achievement Report] [Completion] [Analytics]|
|         |
|         | === Achievement Report Tab === |
|         | Filters: [Dept v] [Quarter v] [Status v] |
|         |
|         | +--Employee+--Dept+--Goal---+Plan+Act+Sc+| | | | | | | |
|         | | Rahul S   |Sales | Revenue |40L |18L|45%||
|         | | Priya M   |Mktg  | Campaigns| 10 | 7 |70%||
|         | +-----+-----+-----+-----+-----+|
|         |
|         | Score Colors: >=80%=Green, 50-79%=Yellow, <50%=Red|
|         |
|         | [Export CSV]   [### Export Excel ###] |
|         |
|         | === Analytics Tab === |
|         | [Bar: Dept scores] [Line: QoQ] [Pie: Status]|
+-----+-----+

```

Screen 13 — Audit Trail Page (Admin)

```

+-----+-----+
| SIDEBAR | Audit Trail — Post Lock Changes |
|         |
|         | Filters: [Employee v] [Date From] [To] |
|         |
|         | +--By-----+--Goal---+--Change-----+--Old+---New+---When--+| | | | | | | |
|         | | Admin    | Revenue | target_change|Rs.40L|Rs.50L| 2 hrs ||
|         | | Manager  | Partners| weightage    | 20%  | 25%  | 1 day ||
|         | | Admin    | Revenue | goal_unlocked|locked|draft | 2 days||
|         | +-----+-----+-----+-----+-----+|
+-----+-----+

```

Screen 14 — Escalation Rules Page (Admin)

```

+-----+-----+
| SIDEBAR | Escalation Configuration |
|         |
|         | Escalation Rules | | | | | |
|         | +--Rule-----+--Days+---On/Off+---Save-+|
|         | | Goal not submitted      | [7 ] | Toggle | Save ||
|         | | Goal not approved       | [5 ] | Toggle | Save ||
|         | | Check-in not completed  | [3 ] | Toggle | Save ||
|         | +-----+-----+-----+-----+|
|         |
|         | Escalation Log | | | | | | |
|         | +--Type-----+--For---+--Against+---Level+---Status-+|
|         | | Goal not submit | Arun K | Manager | 1     | Pending ||

```

```

|           | | Not approved      | Priya M| Rahul S  | 2      | Notified||
|           | +-----+-----+-----+-----+-----+
|           |                                     [Mark Resolved] button per row      |
+-----+-----+-----+-----+-----+

```

6.4 Navigation Flow Diagram

```

LOGIN
|
+---> EMPLOYEE
|     |--- Dashboard (stat cards, goals summary)
|     |--- My Goals --> Create/Edit Goal (modal)
|     +--- Check-ins --> Quarterly check-in form
|
+---> MANAGER
|     |--- Dashboard (team overview)
|     |--- Team Goals --> Goal Review [per employee]
|     |--- Check-in Review --> Review [per employee]
|     +--- Shared Goals --> Push KPI to team
|
+---> ADMIN
|     |--- Dashboard (big picture + activity feed)
|     |--- Cycle Management (open/close windows)
|     |--- User Management (create/assign users)
|     |--- Reports --> Achievement / Completion / Analytics
|     |--- Audit Trail (post-lock change log)
|     +--- Escalations (rules config + log)

```

SECTION 7 — APP ARCHITECTURE

7.1 Complete Folder Structure

```

goal-tracker/
|
+-- app/                                # Next.js App Router
|   +-- (auth)/
|   |   +-- login/page.jsx             # Screen 1 — Login
|   |
|   +-- (dashboard)/
|   |   +-- employee/
|   |   |   +-- layout.jsx             # Employee layout + black sidebar
|   |   |   +-- dashboard/page.jsx     # Screen 2
|   |   |   +-- goals/page.jsx        # Screen 3
|   |   |   +-- checkins/page.jsx     # Screen 5
|   |   |
|   |   +-- manager/
|   |   |   +-- layout.jsx             # Manager layout + sidebar
|   |   |   +-- dashboard/page.jsx     # Screen 6
|   |   |   +-- team-goals/page.jsx    # Screen 7 list
|   |   |   +-- team-goals/[employeeId]/page.jsx # Review
|   |   |   +-- checkin-review/page.jsx # Screen 8 list
|   |   |   +-- checkin-review/[employeeId]/page.jsx
|   |   |   +-- shared-goals/page.jsx  # Screen 9
|   |   |
|   |   +-- admin/
|   |   |   +-- layout.jsx             # Admin layout + sidebar
|   |   |   +-- dashboard/page.jsx     # Screen 10
|   |   |   +-- cycles/page.jsx        # Screen 11
|   |   |   +-- users/page.jsx         # User Management
|   |   |   +-- reports/page.jsx       # Screen 12
|   |   |   +-- audit/page.jsx         # Screen 13
|   |   |   +-- escalations/page.jsx   # Screen 14
|   |   |
|   +-- api/                          # Backend API routes
|   |   +-- auth/route.js
|   |   +-- goals/route.js + [id]/route.js
|   |   +-- checkins/route.js + [id]/route.js
|   |   +-- cycles/route.js + [id]/route.js
|   |   +-- users/route.js + [id]/route.js
|   |   +-- shared-goals/route.js
|   |   +-- audit/route.js
|   |   +-- reports/achievement/route.js
|   |   +-- reports/completion/route.js
|   |   +-- escalations/route.js + rules/route.js
|   |
+-- components/
|   +-- ui/                            # shadcn/ui: Button, Card, Input, Select...
|   +-- layout/                        # Sidebar, Navbar, RoleGuard
|   +-- goals/                        # GoalCard, GoalForm, WeightageBar, GoalTable
|   +-- checkins/                    # CheckInCard, CheckInForm, ProgressScore

```

```

|   +-- dashboard/      # StatCard, DeptProgress, ActivityFeed
|   +-- reports/        # AchievementTable, ExportButton, analytics/
|   +-- escalations/    # EscalationRuleCard, EscalationLog
|
| +-- lib/
| |   +-- supabase/client.js + server.js
| |   +-- utils/scoreCalculator.js
| |   +-- utils/dateHelpers.js
| |   +-- utils/exportHelpers.js
| |   +-- utils/validators.js
| |   +-- utils/escalationChecker.js
| |
| +-- hooks/
| |   +-- useAuth.js, useGoals.js, useCheckIns.js
| |   +-- useCycle.js, useRealtime.js
| |
| +-- middleware.js      # Route protection + role redirects
| +-- tailwind.config.js # Black + yellow theme config
| +-- .env.local         # Supabase URL + anon key

```

7.2 All 20 API Routes

Route	Method	Role	Description
/api/goals	GET	All	Fetch goals filtered by role + active cycle
/api/goals	POST	Employee	Create goal with all validations
/api/goals/[id]	PUT	All	Update goal status, target, weightage
/api/checkins	GET	All	Fetch check-ins filtered by role
/api/checkins	POST	Employee	Submit quarterly check-in
/api/checkins/[id]	PUT	Manager	Add manager comment, mark done
/api/cycles	GET	All	Get active cycle and all window dates
/api/cycles	POST	Admin	Create new yearly cycle
/api/cycles/[id]	PUT	Admin	Update window open/close dates
/api/users	GET	Mgr/Admin	Get team members or all users
/api/users	POST	Admin	Create new user with role assignment
/api/users/[id]	PUT	Admin	Update user role, department, manager
/api/shared-goals	POST	Mgr/Admin	Push shared goal to selected employees
/api/shared-goals	GET	All	Get shared goals for current user
/api/audit	GET	Admin	Get full audit trail with filters
/api/reports/achievement	GET	Admin	Achievement report — all employees
/api/reports/completion	GET	Admin	Check-in completion real-time status

/api/escalations	GET	Admin	Get escalation log
/api/escalations	POST	System	Create new escalation record
/api/escalations/rules	GET+PUT	Admin	Get and update escalation rules

7.3 Three Layers of Security

Layer	Where	What It Does	Protects Against
1 — Middleware	middleware.js (runs on every request)	Redirects unauthenticated users + wrong role	Accidental access, URL guessing
2 — RoleGuard	RoleGuard.jsx (React component)	Hides UI buttons and sections by role	UI confusion, accidental actions
3 — RLS	Supabase Row Level Security (DB level)	Database refuses wrong-role queries	Direct API attacks, bypassed UI

7.4 Data Flow Architecture Diagram

```

=== BROWSER ===

[Employee Pages] [Manager Pages] [Admin Pages]
    |               |               |
    +-----+-----+-----+
    |
    [Custom React Hooks]
    useGoals / useCheckIns
    useCycle / useRealtime
    |
    HTTP requests / fetch()
    |

=== NEXT.JS SERVER ===

    [API Routes - /api/*]
    |
    [Middleware - role checking]
    |
    [Supabase Server Client]
    |

=== SUPABASE ===

[Auth]      [PostgreSQL - 9 tables]      [Realtime]
3 roles      |                          4 tables live
              |
              [DB Triggers]
              score_calc | audit_log | updated_at
              |

```

```
[RLS Policies]
employee | manager | admin rules
```


SECTION 8 — BUILD ORDER (PHASE BY PHASE)

Core Principle: Build vertically — one complete feature at a time. Never build all UI first then all APIs. Always have something demo-able.

Phase 0 — Project Setup

- Create Next.js project, install Tailwind + shadcn/ui + all dependencies
- Configure black (#000000) + yellow (#FDB813) theme in tailwind.config.js
- Create Supabase project, run all 9 table SQL scripts
- Set up RLS policies and 3 DB triggers
- Create .env.local with Supabase credentials
- Seed demo data: 8 thrust areas, 1 active cycle, 7 demo users
- Test Supabase connection from Next.js
- Deploy empty app to Vercel — get live URL immediately

Complete When: Project runs locally, DB ready, live URL exists, GitHub first commit done.

Phase 1 — Authentication

- Build Login page — email + password + Supabase Auth + role-based redirect
- Build middleware.js — protect all routes, redirect by role
- Build Sidebar component (black, role-aware navigation links)
- Build Navbar component (top bar with user info + logout)
- Build 3 empty dashboard shells (just layout, no content yet)

Complete When: All 3 roles log in, land on correct dashboard, sidebar shows correct menu.

Phase 2 — Employee Goal Creation

- Build /api/goals GET + POST routes with all validation
- Build useGoals.js hook
- Build WeightageBar component (live meter updating in real time)
- Build GoalForm component — all 6 fields, UoM changes target input type
- Build GoalCard component with status badges
- Build My Goals page — cards, WeightageBar, Add Goal button, Submit All
- Build Employee Dashboard — 4 stat cards, goals summary table
- Enforce all validations: 100% total, min 10%, max 8 goals

Complete When: Employee creates/submits goals, all validations block wrong inputs, dashboard shows correct stats.

Phase 3 — Manager Approval

- Build `/api/goals/[id]` PUT route (update status, target, weightage)
- Build `/api/users` GET route (manager fetches team)
- Build Manager Dashboard — team overview table, pending count
- Build Team Goals page — list of team members with submission status
- Build Goal Review page per employee — inline edit, approve/return/edit buttons
- Goal locking on full approval, audit log auto-fires on edits

Complete When: Manager approves goals, goals lock, audit log auto-creates, return comment visible to employee.

Phase 4 — Quarterly Check-ins

- Build `/api/checkins` GET + POST + PUT routes
- Build `useCheckIns.js` hook
- Build `lib/utils/scoreCalculator.js` — all 4 UoM formulas
- Build Check-in page — actual achievement inputs, live score preview, notes field
- Build Check-in Review page — planned vs actual table, manager comment form
- Build `useCycle.js` — checks if current window is open, disables inputs if closed

Complete When: Check-in works, scores calculate correctly for all 4 UoM types, manager reviews, window enforcement works.

Phase 5 — Admin Core

- Build `/api/cycles` GET + POST + PUT routes
- Build Cycle Management page — window controls, create cycle form
- Build User Management page — list, create, assign role + manager
- Build Admin Dashboard — big picture stats, dept completion bars, activity feed
- Build Shared Goals flow — create KPI, select recipients, push to team
- Build `/api/shared-goals` routes, test achievement sync

Complete When: Admin controls windows, user management works, shared goals push and sync correctly.

Phase 6 — Reports & Audit

- Build `/api/reports/achievement` route — joined query across 3 tables
- Build `/api/reports/completion` route

- Build Achievement Report table — filters, color-coded scores
- Build ExportButton — CSV and Excel export using SheetJS
- Build Completion Dashboard — Supabase Realtime subscription, live updates
- Build Audit Trail page — /api/audit route, who/what/when table with filters

Complete When: Excel export downloads correctly, completion dashboard updates live, audit trail shows post-lock changes.

Phase 7 — Analytics Module (Bonus)

- Set up Recharts
- Build BarChart — department-wise average scores (black bars, yellow highlights)
- Build LineChart — quarter-on-quarter progress per department
- Build PieChart — goal status distribution
- Add Analytics tab to Reports page, connect all charts to real data

Complete When: 3 charts render with real data, look impressive during demo.

Phase 8 — Escalation Module (Bonus)

- Build Escalation Rules page — threshold editing, on/off toggle per rule
- Build /api/escalations/rules routes
- Build escalationChecker.js — checks all 3 rule types, creates escalation records
- Build Escalation Log table — who/type/level/status, mark resolved button
- Build /api/escalations routes

Complete When: Rules configurable, log shows triggered escalations, can demo with pre-seeded data.

Phase 9 — Polish & Bug Fix

- Add loading skeleton states to all pages
- Add error states with retry button to all pages
- Add empty states with helpful messages + action buttons
- Check mobile responsiveness on all 14 screens
- Consistent black & yellow theme audit across all pages
- Test all 3 roles end to end — full journey per role
- Test all validations: >100% blocked, >8 goals blocked, outside window blocked
- Test edge cases: 0 goals, 0 team members, no active cycle
- Seed realistic demo data — 7 users, realistic goals, Q1 check-ins (some done, some pending)

Complete When: Zero broken flows, all edge cases handled, demo data looks realistic.

Phase 10 — Demo Prep

- Write demo script (which screens in which order — written during Phase 9)
- Note all 7 login credentials
- Practice demo 2-3 times
- Create architecture diagram — show both hackathon and production stacks
- Final Vercel deployment — confirm live URL works
- Submit: GitHub repo, architecture diagram, live URL, login credentials

Complete When: Demo runs smoothly without fumbling, all submission items ready.

Cut Strategy if Running Behind

Situation	What to Cut	What to Keep
Slightly behind	Skip Phase 8 — Escalation Module	Phases 0-7, 9, 10
Further behind	Skip Phase 7 + 8 — both bonus features	Phases 0-6, 9, 10
Very behind	Skip all bonus features, simplify reports	Core Phases 0-6 must be bug-free

NEVER cut Phases 0-6. A bug-free, beautiful, complete core app scores higher than a buggy app with bonus features.

SECTION 9 — AI PROMPTING STRATEGY

9.1 Seven Golden Rules of Vibe Coding

14. Always give CONTEXT before asking to build — paste master context first
15. One feature at a time — never ask for everything at once
16. Always specify the tech stack in every prompt
17. Always reference the design system (black + yellow) explicitly
18. Always tell the AI what NOT to do — prevents unwanted choices
19. When it makes a mistake — correct precisely, not vaguely
20. Always ask it to explain what it built after — catches misunderstandings

9.2 The Master Context Prompt

Save this and paste it at the START of every new AI conversation or when starting a new feature:

```
I am building a Goal Setting & Tracking Portal for a hackathon.
```

```
TECH STACK:
```

- Framework: Next.js 14 (App Router only — never Pages Router)
- Styling: Tailwind CSS + shadcn/ui (never raw CSS)
- Database: Supabase (PostgreSQL + Auth + Realtime)
- Charts: Recharts | Excel: SheetJS | Deployment: Vercel

```
DESIGN SYSTEM:
```

- Primary buttons/accents: #FDB813 (yellow)
- Sidebar/Navbar: #000000 (black)
- Page background: #F5F5F5 | Cards: #FFFFFF
- Never use any other colors

```
THREE ROLES: employee / manager / admin
```

```
KEY BUSINESS RULES:
```

- Total goal weightage per employee = exactly 100%
- Min weightage per goal = 10% | Max goals = 8
- Goals lock after manager approval
- Progress score = calculated by DB trigger ONLY
- Employees can only update during open cycle windows
- Every post-lock change must create an audit log entry

```
9 DB TABLES: users, goals, thrust_areas, cycles,  
check_ins, shared_goals, audit_logs,  
escalation_rules, escalations
```

```
See RULES.md for full context.
```

Now build: [DESCRIBE SPECIFIC FEATURE HERE]

9.3 Key Prompts Per Phase

Phase	What to Say
Phase 0 — Setup	"Set up Next.js 14 with Tailwind, configure #FDB813 as primary color and #000000 as sidebar color. Install shadcn/ui and Supabase. Create folder structure as specified. Do not build any pages yet."
Phase 1 — Auth	"Build the Login page. Email + password form. Connect to Supabase Auth. After login, fetch role from public.users and redirect: employee=/employee/dashboard, manager=/manager/dashboard, admin=/admin/dashboard."
Phase 2 — Goals	"Build the GoalForm component. 6 fields: thrust area dropdown, title, description, UoM dropdown, target (switches to date picker if UoM=timeline), weightage. Show live remaining weightage. Validate min 10%, max 100% total, max 8 goals."
Phase 3 — Approval	"Build the Goal Review page for managers. Show all goals for [employeeId]. Each goal has inline edit for target/weightage, plus Approve/Return/Edit&Approve buttons. On approve, set status=locked. On edit, create audit_log entry."
Phase 4 — Check-in	"Build the Check-in page. For each approved goal, show actual achievement input, status dropdown, notes textarea. Show live score preview using scoreCalculator.js. Only show if active cycle window is open."
Phase 5 — Admin	"Build the Cycle Management page. Show all 5 windows with their dates and current status (Upcoming/Open/Closed). Add Open Now and Close Now buttons that update the dates. Add Create New Cycle form."
Phase 6 — Reports	"Build Achievement Report table with filters for department, quarter, status. Color-code Score column: >=80% green, 50-79% yellow, <50% red. Add Export to Excel button using SheetJS."
Phase 7 — Analytics	"Add Analytics tab to Reports page. Build 3 Recharts charts: BarChart (dept avg scores, #FDB813 bars), LineChart (quarter-on-quarter), PieChart (goal status distribution). Connect to real data."

9.4 Debugging Prompts

When AI builds wrong thing:

"That is not what I asked for. I specifically need [X]. Keep everything else exactly the same. Only change [specific thing]."

When Supabase throws an error:

"I am getting this error: [paste error]. My table structure is: [paste]. My RLS policies are: [paste]. What is wrong? Fix only this issue."

When validation is broken:

"Weightage validation is not working. It must enforce: total = exactly 100% before submit, minimum 10% per goal, maximum 8 goals. Current behavior: [describe]. Fix only the validation, do not change anything else."

When styles are wrong:

"Fix styling only. Primary buttons must use bg-[#FDB813] text-black. Sidebar must use bg-[#000000]. Page background must use bg-[#F5F5F5]. Do not touch any functionality."

9.5 Power Tips for Vibe Coding

Tip	What to Do and Why
Always start session with context	Paste master context first. AI has no memory between sessions — this file IS the memory.
After AI builds something	Ask: "Explain what you built and how it connects to the database schema." Catches misunderstandings early.
One file at a time	Never ask for multiple unrelated files at once. AI quality drops significantly with too many simultaneous concerns.
Save working code immediately	git add . && git commit -m "Phase X done" after every working feature. Never lose working code to AI refactoring.
Stop unsolicited refactoring	If AI says "let me also refactor this for better structure" — say "No. Only add what I asked for." Refactoring breaks working code.
Point to specific rule violations	When AI violates a rule say "You violated Business Rule #3 — max 8 goals. Fix only that." Not "this is wrong fix it."

SECTION 10 — RULES.MD FILE (AI INSTRUCTION FILE)

The RULES.md file is provided as a separate file alongside this document. It is the single source of truth for the AI coding assistant.

10.1 What RULES.md Contains

Section	Contents
Who You Are	AI behavior instructions — how to act, what to prioritize, what to never skip
Tech Stack	Every library with strict do/do-not rules
Design System	Exact hex colors, typography, layout rules — nothing ambiguous
Database Schema	All 9 tables with exact column definitions, types, and constraints
User Roles & Access	What each role can and cannot do in the system
Business Rules	16 numbered non-negotiable rules — AI must never violate these
Progress Score Formulas	Exact JavaScript calculation logic for all 4 UoM types
Folder Structure	Complete file tree the AI must follow
API Design Rules	Consistent response format, HTTP status codes, query efficiency
Performance Rules	Caching strategy, N+1 prevention, realtime configuration
Component Requirements	Every component must have loading/error/empty states
Things to Never Do	16 hard restrictions the AI must never violate
Demo Credentials	All 7 test accounts for judges
Build Order	Phase by phase with cut rules if running behind
Testing Checklist	Per-phase verification checklist
How to Use	Instructions for any AI IDE or chat-based AI tool

10.2 How to Use RULES.md

- In any AI-powered IDE: Add as project rules / system prompt / custom instructions file
- In chat-based AI: Paste entire file at the very start of every new conversation
- When starting a new chat: Always paste RULES.md first — AI has no memory between sessions
- When AI violates a rule: Point to the specific numbered rule e.g. "You violated Business Rule #3"
- When context is lost in long conversation: Re-paste RULES.md and continue

SECTION 11 — DEMO CREDENTIALS & SUBMISSION CHECKLIST

11.1 All 7 Demo User Accounts

Role	Email	Password	Department	Reports To
Admin	admin@goaltrack.com	Admin@123	HR	—
Manager 1	manager1@goaltrack.com	Manager@123	Sales	Admin
Manager 2	manager2@goaltrack.com	Manager@123	R&D	Admin
Employee 1	emp1@goaltrack.com	Employee@123	Sales	Manager 1
Employee 2	emp2@goaltrack.com	Employee@123	Sales	Manager 1
Employee 3	emp3@goaltrack.com	Employee@123	R&D	Manager 2
Employee 4	emp4@goaltrack.com	Employee@123	R&D	Manager 2

11.2 Mandatory Submission Items

- Live hosted demo URL (Vercel — get this in Phase 0 immediately)
- GitHub / GitLab repository with all commits (version history matters)
- Architecture diagram PDF or image — show BOTH hackathon stack and production Azure path
- Login credentials for all 3 roles (or provide all 7 accounts above)

11.3 When to Write the Demo Script

Write the demo script during Phase 9 (polish & bug fix) when the app is nearly complete — not before. Base it on what actually works, not assumptions.

The demo script will cover:

- Opening pitch — 30 seconds: what problem, why it matters, one-line solution
- Screen-by-screen flow — 8-10 minutes: scripted journey through all 3 roles
- Talking points per screen — what to say while clicking each feature
- Closing statement — 30 seconds: architecture summary, cost comparison, why this beats Lattice
- Backup plans — what to skip if something breaks during live demo
- Anticipated judge questions with prepared answers

SECTION 12 — COMPLETE TESTING CHECKLIST

Phase 1 — Authentication

- All 3 roles can log in successfully
- Wrong password shows clear error message
- Each role lands on their correct dashboard after login
- Sidebar shows correct navigation links per role
- Employee accessing /manager/* gets redirected to employee dashboard
- Manager accessing /admin/* gets redirected to manager dashboard

Phase 2 — Employee Goal Creation

- Employee can create a goal with all 6 fields
- Thrust area dropdown shows only active thrust areas
- UoM dropdown changes target input (number vs date picker)
- WeightageBar updates in real time as employee types weightage
- Adding goal below 10% weightage shows error and is blocked
- Adding more than 8 goals shows error and blocks the 9th
- Cannot submit for approval if total weightage is not exactly 100%
- Submit All button changes all draft goals to submitted status
- Goal Setting window closed => Add Goal button disabled with message
- Employee dashboard shows correct stat card counts

Phase 3 — Manager Approval

- Manager Dashboard shows only their direct reports
- Pending approval count is accurate
- Manager can open any employee goal sheet
- Can edit target inline => goal saves with new target
- After edit => audit_logs table has new entry with old/new values
- Can approve single goal => status changes to locked
- Can return goal with comment => employee sees comment on their goal card
- Approve All button locks all goals for employee at once
- After all approved => employee cannot edit any goals

Phase 4 — Quarterly Check-ins

- Q1 window open => Employee sees check-in form for all approved goals
- Q1 window closed => Check-in form hidden with message showing next window date
- numeric_min: 18L actual / 40L target = 45% score (verify in DB)
- numeric_max: 3hrs actual, 5hrs target = 100% (capped) score
- timeline: actual_date = target_date = 100% score
- timeline: actual_date > target_date = 0% score
- zero_based: actual = 0 = 100% score
- zero_based: actual = 1 = 0% score
- Cannot submit check-in without entering actual values for all goals
- Manager can see Planned vs Actual table for each employee
- Manager can add comment and mark check-in done
- Completion dashboard shows this employee as done

Phase 5 — Admin Core

- Admin can open Q1 window => employees immediately see check-in available
- Admin can close Q1 window => check-in immediately disabled for employees
- Admin can create a new user with any role
- Shared goal pushed to 3 employees => all 3 see it in their goals
- Recipients cannot edit title or target of shared goal
- Recipients can edit weightage of shared goal
- Primary owner updates achievement => all 3 linked employees get same score

Phase 6 — Reports & Audit

- Achievement report shows all employees in the system
- Department filter correctly filters rows
- Quarter filter shows correct check-in data
- Score color coding works: >=80% green, 50-79% yellow, <50% red
- Export CSV downloads a valid CSV file with all filtered data
- Export Excel downloads a valid .xlsx file with all filtered data
- Completion dashboard shows real-time who has/has not done check-ins
- Completing a check-in immediately updates the completion dashboard (Realtime)
- Audit trail shows all post-lock changes with who/what/when
- Audit trail date filter correctly filters entries

SECTION 13 — QUICK REFERENCE CARD

13.1 The 16 Non-Negotiable Business Rules

#	Rule	Enforced By
1	Total weightage = exactly 100% before submission	Frontend validation + API
2	Each goal minimum 10% weightage	Frontend + DB CHECK constraint
3	Maximum 8 goals per employee per cycle	API route validation
4	Goals are LOCKED after manager approval	Status machine in API
5	Locked goals CANNOT be edited without admin	RLS + API status check
6	Every post-lock change MUST create audit log	DB trigger (automatic)
7	Shared goal title and target are READ-ONLY	UI disabled + API validation
8	Recipients can ONLY change weightage of shared goal	UI + API permission check
9	Primary owner achievement SYNCS to linked employees	API logic on check-in POST
10	Employees can ONLY check-in during open windows	useCycle hook + API date check
11	One check-in per goal per quarter per cycle	DB UNIQUE constraint
12	Progress score AUTO-CALCULATED by DB trigger ONLY	Supabase PostgreSQL trigger
13	Only ONE cycle can be active at a time	Partial unique index + API
14	Admin controls all window open/close dates	Role check in cycle API
15	Outside window => all relevant inputs DISABLED	useCycle hook in UI
16	Escalation N-day threshold CONFIGURABLE by admin	Escalation rules table

13.2 Score Calculation Reference

UoM Type	Formula	Example
numeric_min	$\text{MIN}(\text{actual} / \text{target} * 100, 100)$	$18\text{L}/40\text{L} = 45\%$
numeric_max	$\text{MIN}(\text{target} / \text{actual} * 100, 100)$	$5/3 = 100\%$ (capped)
timeline	$\text{actual} \leq \text{target} ? 100 : 0$	On time = 100%
zero_based	$\text{actual} === 0 ? 100 : 0$	0 incidents = 100%

Always round to 2 decimal places.

```
Always handle division by zero gracefully (return 0).
```

13.3 Things to NEVER Do

#	Never Do This	Do This Instead
1	Never store raw passwords	Use Supabase Auth — it handles hashing
2	Never calculate progress score on frontend and store it	DB trigger auto-calculates — trust it
3	Never allow editing locked goals from UI	Check status === locked before showing edit button
4	Never allow check-in outside open window	useCycle hook gates all check-in inputs
5	Never allow more than 8 goals	Check count before allowing POST /api/goals
6	Never submit if total weightage is not 100%	Disable Submit All button until WeightageBar = 100%
7	Never allow goal weightage below 10%	Real-time validation + DB CHECK constraint
8	Never use colors outside design system	Only use #FDB813, #000000, #F5F5F5, #FFFFFF + status colors
9	Never use Pages Router	Always App Router — folder structure is app/
10	Never write inline styles	Always Tailwind CSS utility classes
11	Never skip error handling in API routes	Always wrap in try/catch, return proper status codes
12	Never expose Supabase service role key to browser	Only use service key in server-side API routes
13	Never do SELECT * queries	Always select only needed columns for performance
14	Never skip loading/error/empty states	Every component must handle all 3 states
15	Never do N+1 database queries	Always use joins to fetch related data in one query
16	Never install libraries outside tech stack	Ask before adding anything new

13.4 Caching Strategy Reference

Data Type	Cache Duration	Next.js Setting
Thrust areas list	Forever (static)	next: { revalidate: false }
Active cycle dates	1 hour	next: { revalidate: 3600 }

Employee goals	5 minutes	next: { revalidate: 300 }
Manager team list	30 minutes	next: { revalidate: 1800 }
Completion dashboard	Never (realtime)	cache: 'no-store'
Progress scores	5 minutes	next: { revalidate: 300 }

GoalTrack — AtomQuest Hackathon 1.0

Complete Project Planning Document | Version 1.0

Build order: Setup -> Auth -> Goals -> Approval -> Check-ins -> Admin -> Reports -> Analytics -> Escalation -> Polish
-> Demo